

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Mechanical Engineering

2. Subject Name: Design of Machine Elements L-T-P: 3-1-0 Credits: 4

3. Pre-requisites:

4. Syllabus and reference books:

Syllabus:

Material Selection: Ashby charts.

Design for Strength: Theories of failure for brittle and ductile materials under static loading.

Fatigue: Types of fatigue loading and failure, Fatigue test, endurance limit; Fatigue under combine stresses; Factors influencing fatigue behaviour.

Allowable stress, factor of safety.

Introduction to bearings and selection of bearings. Design of Shafts, keys and keyways.

Review of Basic of gear nomenclature of spur, helical; design of Spur gears for simple power transmission, Design of Springs,

Design of Clutches and Brakes, Design of belt drives, Design of fasteners, rivets, and-welded joints

Reference Books: Shigley's Mechanical Engineering Design by Richard G. Budynas and J. Keith Nisbett